FINITENESS APPEARS AT A FINITE STAGE UNDER FINITE TRANSITIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1606-1614, following the statement at 1591-1604.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Integral morphisms are etale-locally detected. Morphisms of Spaces, Lemma spaces-morphisms-lemma-integral-local (cross-chapter, cited at ~1610). It suffices to check X_i x_Y W -> W finite after a surjective etale W -> Y.
- Limits that are schemes. lemma-limit-is-scheme (same chapter, cited at ~1611). Reduces to the case of schemes.
- Scheme-level eventual finiteness. Limits, Lemma limits-lemma-eventually-finite (cross-chapter, cited at ~1612-1613). Concluding engine for schemes.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A0S: focused-seeds/geometry/S9-spaces-limits-eventually-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZQ: focused-seeds/geometry/S9-spaces-limits-eventually-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let Y be an algebraic space over S. Let $X = \lim X_i$ be a directed limit of algebraic spaces over Y with affine transition morphisms. Assume*

(1) *Y quasi-compact and quasi-separated,*
(2) *X_i quasi-compact and quasi-separated,*
(3) *the transition morphisms $X_{i'} \to X_i$ are finite,*
(4) *$X_i \to Y$ locally of finite type*
(5) *$X \to Y$ integral.*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Then $X_i \to Y$ is finite for i large enough.

Proof. Choose an affine scheme W and a surjective étale morphism $W \to Y$. Then $X \times_Y W$ is finite over W and it suffices to check that $X_i \times_Y W$ is finite over W for some i (Morphisms of Spaces, Lemma 03ZQ). By Lemma 07SR this reduces us to the case of schemes. In the case of schemes it follows from Limits, Lemma 0A0S. □